The C.F. is $(Ax+B)e^{\frac{2}{3}x}$

The general solution is $y = (Ax+B)e^{\frac{2}{3}x}$

Example 4.27

Solve $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 5y = 0$

Solution:

Given $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 5y = 0$

$$(D^2 - 4D + 5)y = 0$$

The auxiliary equation is

$$m^2 - 4m + 5 = 0$$

$$\Rightarrow (m-2)^2 - 4 + 5 = 0$$

$$(m-2)^2 = -1$$

$$m - 2 = \pm\sqrt{-1}$$

$m = 2 \pm i$, it is if the form $\alpha \pm i\beta$

$\therefore$ C.F= $e^{2x}\left[A\cos x + B\sin x\right]$

The general solution is

$$y = e^{2x}\left[A\cos x + B\sin x\right]$$

Example 4.28

Solve $\frac{d^2x}{dt^2} - \frac{3dx}{dt} + 2x = 0$ given that when $t = 0,\ x = 0$ *and* $\frac{dx}{dt} = 1$

Solution:

$$\frac{d^2x}{dt^2} - 3\frac{dx}{dt} + 2x = 0$$

Given $(D^2 - 3D + 2)x = 0$ where $D = \frac{d}{dt}$

A.E is $m^2 - 3m + 2 = 0$

$$(m-1)(m-2) = 0$$

$$m = 1, 2$$

$$\text{C.F} = Ae^t + Be^{2t}$$

The general solution is $x = Ae^t + Be^{2t}$ (1)

Now when $t = 0,\ x = 0$ (given)

$$(1) \Rightarrow 0 = A + B \quad (2)$$

Differentiating (1) w.r.t 't'

$$\frac{dx}{dt} = Ae^t + 2Be^{2t}$$

When t = 0, $\frac{dx}{dt} = 1$

$$A + 2B = 1 \quad (3)$$

Thus we have $A + B = 0$ and $A + 2B = 1$

Solving, we get A = –1, B = 1

$\therefore$ (1) $\Rightarrow$ $x = -e^t + e^{2t}$

(i.e.) $x = e^{2t} - e^t$

Type II : $f(x) = e^{ax}\ (i.e)\ \phi(D)y = e^{ax}$

$$\text{P.I} = \frac{1}{\phi(D)}e^{ax}$$

Replace D by a, provided $\phi(D) \neq 0$ when D = a

If $\phi(D) = 0$ when D = a, then

$$\text{P.I} = x\frac{1}{\phi'(D)}e^{ax}$$

Replace D by a, provided $\phi'(D) \neq 0$ when D = a

If $\phi'(D) = 0$ when D = a, then

$$\text{P.I} = x^2\frac{1}{\phi''(D)}e^{ax}$$ and so on

Example 4.29

Solve : $(D^2 - 4D - 1)y = e^{-3x}$

Solution:

$$(D^2 - 4D - 1)y = e^{-3x}$$

The auxiliary equation is

XII Std - Business Maths & Stat EM Chapter 4.indd 96 2/28/2020 6:03:02 PM